Huawei Access Node Optical Transport Failure SOP (JAY-ACC-68)

Node: JAY-ACC-68 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4977

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a Major optical transport failure on Huawei Access Node JAY-ACC-68 located at Jayanagar Metro Site, where the failure domain is related to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. The underlying protocol and hardware mechanics responsible for this failure domain are the Optical Transport Network (OTN) and the Wavelength Division Multiplexing (WDM) technology employed by the Huawei Access Node. The OTN is a transport network architecture that enables the efficient and reliable transmission of client signals over optical fibers, while the WDM technology allows multiple client signals to be transmitted over a single optical fiber, each at a different wavelength. The Huawei Access Node uses the OTN and WDM technology to provide high-speed and reliable data transmission services. However, the degradation of the SFP/XFP, fiber cuts, or Rx/Tx power loss can lead to optical transport failures, causing service disruptions and affecting the quality of service. This SOP provides a step-by-step guide to diagnose and resolve such failures, ensuring minimal downtime and maintaining the reliability of the network.

Pre-requisite Checks

- Verify that the failure is related to the Access Node JAY-ACC-68 at Jayanagar Metro Site.
- Check the redundant link status to ensure that it is functioning properly.
- Dispatch remote hands to the site to perform physical checks and ensure that the site is secure.
- Verify that the Huawei Access Node is powered on and in a stable state.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optics transceiver JAY-ACC-68, Expected Output: Optics Transceiver Information: SFP/XFP status, wavelength, and optical power, Description: This command displays the optics transceiver information, including the SFP/XFP status, wavelength, and optical power. It helps to identify any degradation or malfunction of the SFP/XFP.
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```
Command: display optics fiber JAY-ACC-68, Expected Output: Fiber Information: fiber status, cut-off wavelength, and attenuation, Description: This command displays the fiber information, including the fiber status, cut-off wavelength, and attenuation. It helps to identify any fiber cuts or degradation.

Command: display optics rxtx JAY-ACC-68, Expected Output: Rx/Tx Power Information: received and transmitted power, Description: This command displays the Rx/Tx power information, including the received and transmitted power. It helps to identify any Rx/Tx power loss.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP, Command: replace optics transceiver JAY-ACC-68, Description: This command replaces the SFP/XFP, which may have degraded or malfunctioned, causing the optical transport failure.

Step 2: Step: Repair or replace the fiber, Command: replace optics fiber JAY-ACC-68, Description: This command repairs or replaces the fiber, which may have been cut or degraded, causing the optical transport failure.

Step 3: Step: Adjust the Rx/Tx power, Command: adjust optics rxtx JAY-ACC-68, Description: This command adjusts the Rx/Tx power, which may have been lost due to a malfunction or degradation of the Rx/Tx components.

Post-Check Verification

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Command: display optics transceiver JAY-ACC-68, Expected Output: Optics Transceiver Information: SFP/XFP status, wavelength, and optical power, Description: This command verifies that the SFP/XFP is functioning properly and that the optical power is within the normal range.

Command: display optics fiber JAY-ACC-68, Expected Output: Fiber Information: fiber status, cut-off wavelength, and attenuation, Description: This command verifies that the fiber is functioning properly and that the cut-off wavelength and attenuation are within the normal range.

Command: display optics rxtx JAY-ACC-68, Expected Output: Rx/Tx Power Information: received and transmitted power, Description: This command verifies that the Rx/Tx power is within the normal range and that there is no power loss.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: restore optics transceiver JAY-ACC-68, Description: This command restores the original SFP/XFP, which was replaced during the resolution process.

- Step: Restore the original fiber, Command: restore optics fiber JAY-ACC-68, Description: This command restores the original fiber, which was repaired or replaced during the resolution process.

- Step: Restore the original Rx/Tx power, Command: restore optics rxtx JAY-ACC-68, Description: This command restores the original Rx/Tx power, which was adjusted during the resolution process.